

WEEK-2 PROJECT – Retail Bicycle Analysis

Final Summary

1. Cleaning problems found

Transaction Table-

The dataset has 20k rows and 12 columns. The data looked messy because of unstructured column sizes. The data did not have any spell error or case inconsistencies. There were no duplicate values found. Date format was correct.

Customer Demographic-

No duplicate values found. Converted date format, from custom to short date. In Gender columns contains abbreviation like F, Femal ,Female, M and Male. 1 replacement made (F -> Female). 1 replacement made (Femal -> Female). 1 replacement made (M -> Male).

Customer Address-

No duplicate values. 86 replacements made (New South Wales -> NSW). 82 replacements made (Victoria -> VIC)

2. How they merged sheets

- All the rows from Transaction table were brought to Master Table.
- Customer Demographic and Customer Address data was merged using customer_id.
- Excel function XLOOKUP was used to combine datasets.

The final Master Table contained all required fields including:

- Customer demographics
- Customer Address
- Transaction values
- Revenue and profit data

3. Dashboard

KPI :

- Total Revenue - \$ 2,17,39,495.29
- Total Customers - 3493
- Avg. Order Value - 1107.74
- Top State - NSW
- Top Brand - Solex Bicycles
- Top Wealth Segment - Mass Customer
- Top Industry Category – Manufacturing

Visualizations Included

- Monthly Revenue Trend (Line Chart)
- Revenue by Wealth Segment (Column Chart)
- Revenue by State (Column Chart)
- Revenue by Gender (Column Chart)

Filters (Slicers)

To make the dashboard interactive, slicers were added for:

- State
- Wealth Segment
- Gender

4. Top 5 insights

1. Mass Customer segment generates the highest revenue → Focus marketing campaigns and loyalty programs on mass-market customers.
2. Customers aged 41–50 generate the highest revenue → Target this age group with fitness-focused bicycle promotions.
3. NSW contributes the highest revenue among all states → Increase inventory and regional promotions in NSW to maximize sales.
4. Solex Bicycles generates the highest brand revenue → Strengthen partnerships and promote Solex as a flagship brand.
5. Revenue peaks around August and October → Plan seasonal marketing campaigns to maximize sales.

5. 2 business recommendations

- **Strengthen Partnerships with Top Bicycle Brands**
Brands like Solex Bicycles, WearerA2B, and Giant Bicycles generate the highest sales. The business should collaborate with these brands for exclusive promotions.
- **Exclusive offers for High Net Worth and Affluent customers**
These customers contribute strong revenue despite smaller numbers. Introduce premium bicycle models and exclusive offers for these segments.